Chapter 4

Extent (How Much) Decisions

Learning Objectives

- 4.1 Differentiate between average and marginal costs.
- 4.2 Compute average cost (AC), marginal cost (MC), and marginal revenue (MR).
- 4.3 Identify MR and MC as the relevant costs of an extent decision.
- 4.4 Identify the effect of incentive compensation and fixed fees on effort.

Chapter 4 – Main Points

- Do not confuse average and marginal costs.
- **Average cost** (AC) is total cost (fixed and variable) divided by total units produced.
- Average cost is irrelevant to an extent decision.
- **Marginal cost** (MC) is the additional cost incurred by producing and selling one more unit.
- **Marginal revenue** (MR) is the additional revenue gained from selling one more unit.
- Sell more if MR > MC; sell less if MR < MC. If MR = MC, you are selling the right amount (maximizing profit).
- The relevant costs and benefits of an extent decision are marginal costs and marginal revenue. If the marginal revenue of an activity is larger than the marginal cost, then do more of it.
- An incentive compensation scheme that increases marginal revenue or reduces marginal cost will increase effort. Fixed fees have no effects on effort.
- A good incentive compensation scheme links pay to performance measures that reflect effort.

Warm-Up

- Imagine you're running a food truck. You've already paid to rent the truck and buy equipment. Now you're deciding how long to stay open today. At what point does it stop being worth it to make another sale?

Discussion Activity 1

- Should Troy reopen a parking lot even though the *average cost* per spot is higher than the *price*?

Extent (How Much) Decisions

- When making extent decisions we are concerned with increasing at the margin (marginal benefit vs. marginal cost & marginal revenue vs. marginal cost).
  - Recall that a marginal change is an incremental or additional change.

- Suppose, Troy University is contemplating reopening an old parking lot. If the parking lot will allow 500 additional students to purchase parking passes each year, then Troy gains extra revenue. At the same time, Troy University is currently losing money on all it's parking lots.
- Is it smart for Troy to reopen the parking lot?
- Is Troy facing a sunk cost or hidden cost fallacy?

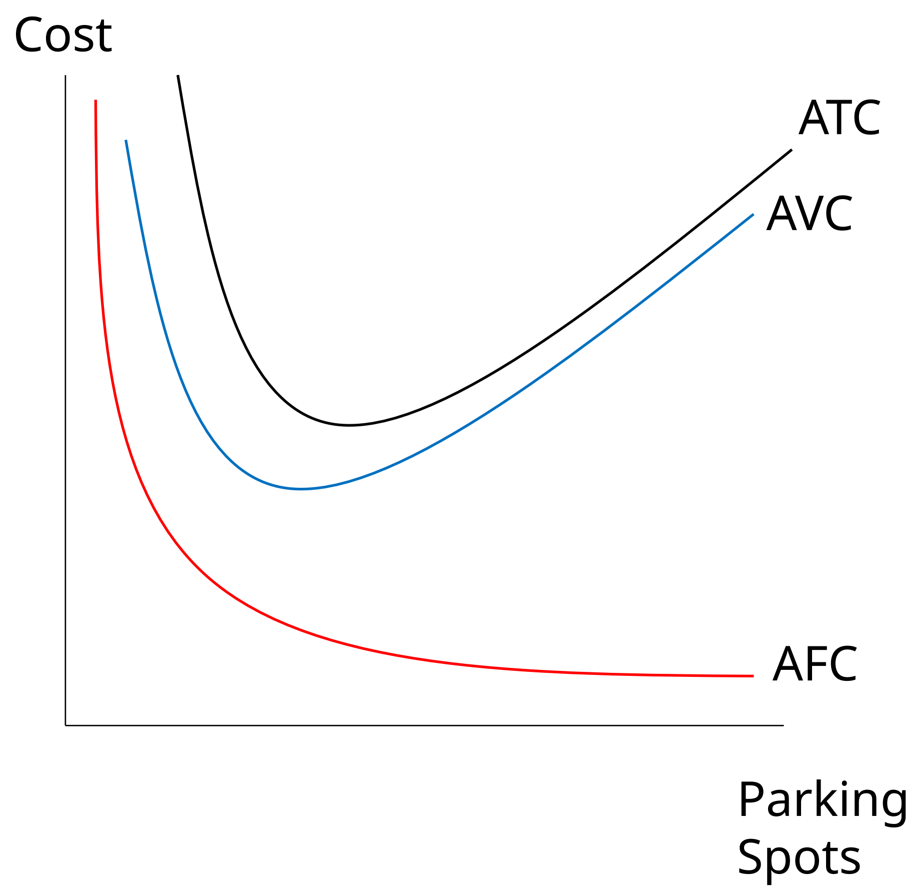
Average Costs

- Troy University currently has 4,000 parking spots available on campus. Each spot generates $75. Each spot also costs the university $81 to maintain each year.

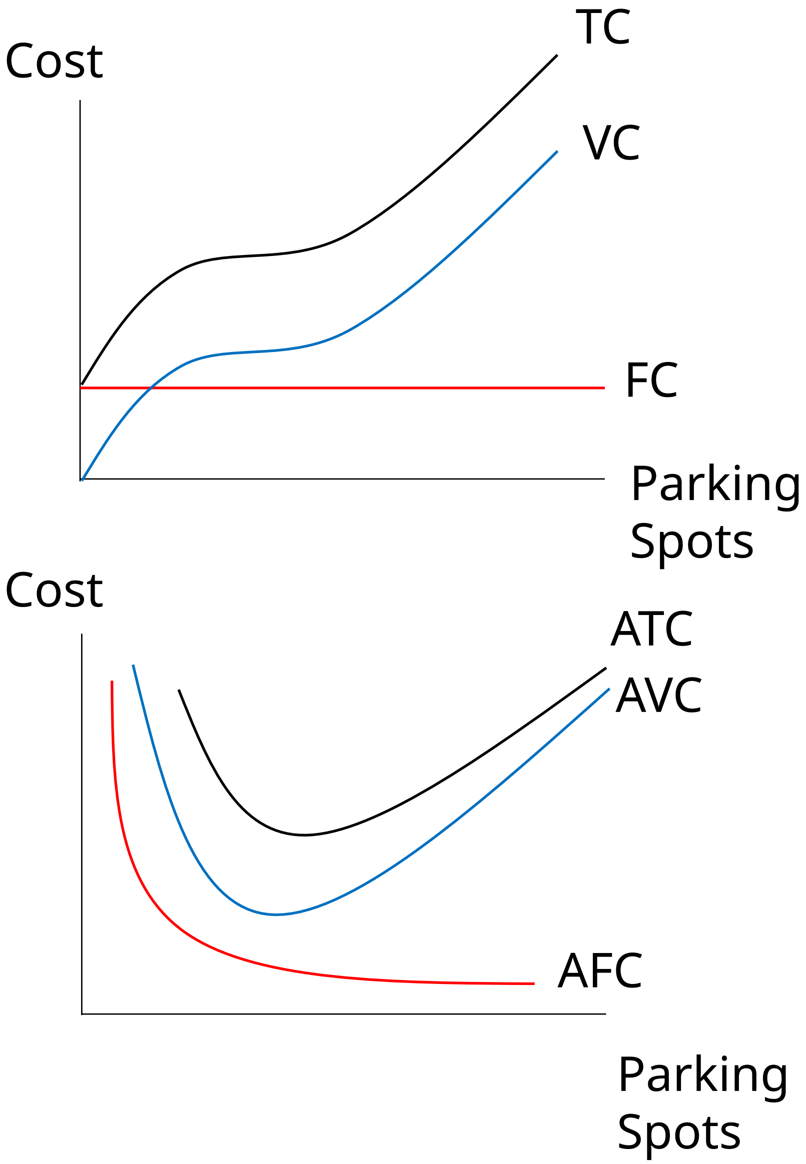
- There are three types of average cost:
  - Average Fixed Cost (AFC) = FC / Q
  - Average Variable Cost (AVC) = VC / Q
  - Average Total Cost (ATC) = TC / Q (or AFC + AVC)

- Because output initially expands faster than costs rise, average variable and average total costs will decrease at first then increase later (law of diminishing returns)
- Fixed costs never change, so average fixed costs will continually decrease as output increases.

Troy University's Average Cost Curves



Comparison of Cost Curves



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Troy’s Parking Problems Example

- Troy University currently has 4,000 parking spots available on campus. Each spot generates $75. Each spot also costs the university an additional $6 to maintain each year.

- If it costs Troy University $300,000 in fixed costs per year to maintain their parking spots, what is Troy’s average fixed cost of parking spots?

AFC = FC/Q
$300,000/4,000 = $75

- What is Troy’s average variable cost for parking spots?

AVC = VC/Q
$24,000/4,000 = $6

- What is Troy’s average total cost for parking spots?

ATC = TC/Q or AFC + AVC
$324,000/4,000 = $81

Marginal Analysis

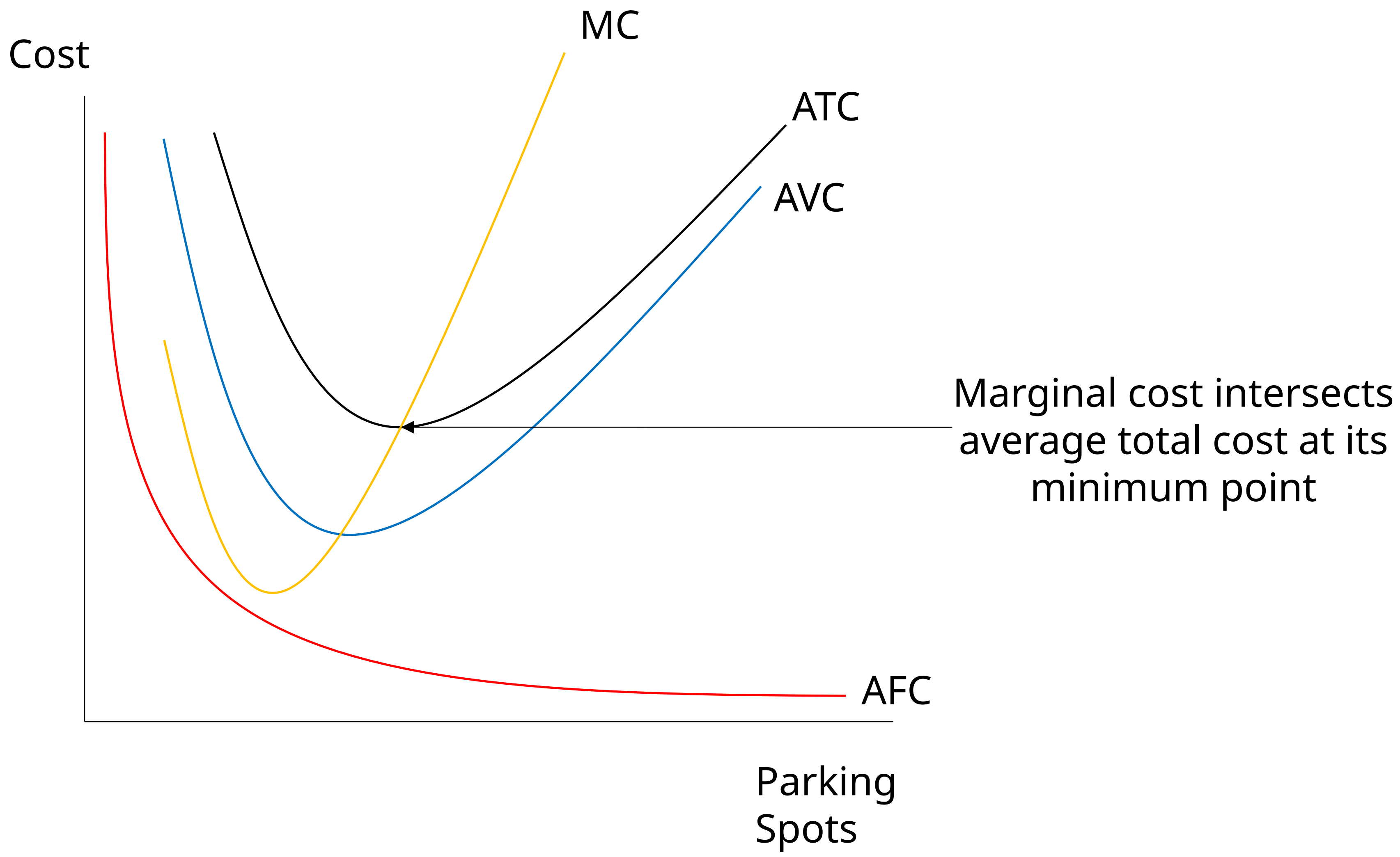
- Troy University may incorrectly decide to not reopen the parking lot because they are currently losing money. Instead, they should focus on marginal analysis.

- Marginal cost – the additional cost incurred by producing and selling one more unit

- Marginal revenue – the additional revenue gain from selling one more unit.

- MR > MC then sell more to increase your total profit
- MR < MC then sell less to increase your total profit
- MR = MC then continue to sell what you are currently selling, as profits are maximized

Introducing the Marginal Cost Curve



Troy's Marginal Cost Example

Number of Spots	Total Cost
0	$300,000
1	$300,006
2	$300,012
3	$300,018
4	$300,024
5	$300,030

The marginal cost of each parking spot is:

MC = $6

The fixed cost of all parking spots are:

FC = $300,000

By adding the 5th spot, Troy incurs how much in additional costs?

MC = $6

Does Troy reopen the parking lot?

- Compare the marginal revenue of the additional spots to the marginal cost of the additional spots.
- Marginal revenue (MR) = 500 * $75 = $37,500
- Marginal cost (MC) = 500 * $6 = $3,000
- Since MR > MC, then yes, Troy should reopen the parking spots. This actually helps Troy increase its revenues

- Total revenue = 4,500 * $75 = $337,500
- Total cost = $324,000 + $3,000 = $327,000
- Profit = $10,500

AC vs. MC – What’s the Difference?

Feature	Average Cost (AC)	Marginal Cost (MC)
What it shows	Cost per unit overall	Cost of *one more* unit
Includes fixed costs?	Yes	No
Used in decisions?	No (not marginal decisions)	Yes
Example	$81 average per parking spot	$6 cost for one more spot

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Activity – Marginal Analysis

Scenario:

- A company sells widgets for $20 each. Producing one more widget costs $15.

1. What is the marginal revenue?
2. What is the marginal cost?
3. Should the company produce another widget?
4. Now, suppose the price drops to $14. What should the company do?

Activity – Marginal Analysis

1. MR = $20
2. MC = $15
3. Since MR > MC, they should produce another unit.
4. If MR = $14 < MC = $15 → they should not produce the extra unit.

Discussion Activity 2

- Why is it a mistake to use average cost when deciding whether to expand output or reopen a service?
- What does average cost include that marginal cost doesn't? Which cost actually changes when output increases?

Activity – Hiring Decisions and Costs

- An additional worker can increase total productivity by 500 units a week. The cost of hiring the worker is $1000 per week. If the price of each unit sold is $1.50, the firm should:
- Suppose the price per unit is $3.00. What should the firm do?
- At 10 units of output, average fixed cost is $50 and average variable cost is $30, then total cost at this output level is:

Activity – Hiring Decisions and Costs

1. Not hire the worker. The marginal costs > marginal revenue
   - MR = 500 x $1.50 = $750
   - MC = $1,000

2. Hire the worker. The marginal revenue > marginal cost
   - MR = 500 x $3.00 = $1500
   - MC = $1,000

3. AFC = $50 and AVC = $30, so ATC = $80
   - ATC = TC/Q
   - $80 = TC/10
   - TC = $800

Activity – Profit Maximization

# Units Produced	Total Revenue	Total Costs
0	0	100
1	200	200
2	350	270
3	440	360
4	520	460
5	590	580

1. What is marg
2. What is marginal cost when output equals 2?
3. When is profit maximized?

Activity – Profit Maximization

# Units Produced	Total Revenue	Total Costs	Marginal Revenue	Marginal Cost
0	0	100	0	0
1	200	200	200	100
2	350	270	150	70
3	440	360	90	90
4	520	460	80	100
5	590	580	70	120

Cost Equations (not on quizzes or tests)

- Many times, we are faced with equations instead of tables. These equations help us estimate various levels of output that minimize cost.

- For example, suppose we have the following:

Total cost = $200 + 75Q - 1.5Q^2 + 0.01Q^3$

1. Determine the output level at which MC is minimized

2. Determine the output level at which AVC is minimized

1. Determine the output level at which MC is minimized

Marginal cost is minimized where it is equal to 0.

$$TC = 200 + 75Q - 1.5Q^2 + 0.01Q^3$$

$$\frac{dTC}{dQ} = 75 - 3Q + 0.03Q^2$$

$$\frac{d^2TC}{dQ^2} = -3 + 0.06Q = 0$$

$$Q = 50$$

2. Determine the output level at which AVC is minimized

AVC is minimized where AVC and MC equal

$$TC = 200 + 75Q - 1.5Q^2 + 0.01Q^3$$

$$AVC = 75 - 1.5Q + 0.01Q^2 \quad = \quad \frac{dTC}{dQ} = 75 - 3Q + 0.03Q^2$$

$$\frac{dAVC}{dQ} = -1.5 + 0.02Q = 0$$

$$Q = 75$$

Average Cost vs. Marginal Cost

	Average Cost (AC)	Marginal Cost (MC)
What is it?	Total cost ÷ total output	Cost of *one more* unit
Based on…	All units made so far	Only the next unit
Used for…	Long-run planning, pricing floor	**Extent decisions** (how much to produce)
Can it mislead?	Yes. It may hide rising costs at the margin	No. It shows whether additional output adds value
Example	$500 for 10 units = $50 AC	11th unit costs $70 → MC = $70

Incentive Pay

- How do we determine the level and type of pay for employees?
- We, again, look at the marginal benefits and marginal costs.
  - If an employee is hard-working and driven by the opportunity to earn a higher salary, then offering some type of incentive scheme would drive sales and increase their pay.
  - On the other hand, if the employee isn't that hard-working, then a flat salary might be the best plan.

- Consider two compensation schemes:
  - Scheme 1 offers 10% commission on all sales, but no flat salary
  - Scheme 2 offers 5% commission on all sales with a $25,000 flat salary
  - Which one is the better alternative?
    - Depends on the employee's willingness to work.

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Incentive Pay

- Suppose the employee expects to sell 100 units of the good and each unit costs $5,000.
- **Scheme 1**: the employee generates $500 from each unit sold for a total of $50,000.
- **Scheme 2**: the employee generates $250 from each unit sold for a total of $50,000 ($25,000 from sales and $25,000 from the flat salary).

- A problem that arises is the difficulty of selling the good.
  - Easier sales require less effort and will be made under each scheme.
  - Difficult sales require more effort and will only be made under the 10% scheme (scheme 1).
  - The employee will tend to self-select themselves into a scheme based on their willingness to work. (More on this in chapter 20)

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Discussion Activity 3

- If you were an employee, which incentive scheme would you choose: high commission and no base pay, or lower commission with a base salary? Why?
- Think about your motivation and confidence in making sales. Also consider risk preferences.

Case Study: Amazon's Dynamic Warehouse Staffing

- Amazon runs massive fulfillment centers across the world. To meet fluctuating demand, especially during peak times (like Prime Day or holidays), Amazon:
  - Offers shift bonuses or surge pay to get workers to take on extra hours.
  - Uses algorithms to calculate when additional workers are needed, and how much they must be paid to bring them in.
  - Sometimes offers higher pay for less desirable shifts (e.g., night or weekend) to encourage sign-ups.

1. **What's the marginal cost of an extra worker or hour?**
   - The bonus or wage paid for each additional shift.
2. **What's the marginal revenue?**
   - The value of orders fulfilled (keeping customers happy and meeting deadlines).
3. **How is incentive pay structured?**
   - Time-based bonuses and dynamic pay create a marginal incentive: work more only when it's profitable for Amazon.
4. **Why not just pay everyone more all the time?**
   - That would raise average cost, not marginal cost, and reduce flexibility.

How does Amazon use marginal thinking in staffing?
Is this system fair to workers? Is it efficient?
Would fixed salaries reduce or increase performance?

Key Takeaways

- Extent decisions ask: *How much more (or less) should we do?*
  → Focus on marginal cost (MC) and marginal revenue (MR).
- Key rule:
  - If MR > MC, do more.
  - If MR < MC, do less.
  - If MR = MC, you're at optimal output.
- Average cost (AC) is irrelevant to extent decisions; it includes fixed costs that don't change with output.
- Fixed costs don't affect marginal analysis; ignore them in short-run decisions.
- Law of diminishing returns: As inputs increase, marginal gains eventually decrease.
- Use marginal thinking to guide hiring, pricing, production, and operational decisions.
- Incentive compensation matters:
  - Fixed salaries don't affect marginal effort.
  - Commission-based pay (or bonuses) can increase effort, if tied to performance.
- Good incentive systems match pay with effort and output, encouraging employees to act in the firm's interest.

Knowledge Check

1. What’s the difference between marginal cost and average cost?
2. When is profit maximized in terms of MR and MC?
3. What does it mean if MR < MC?
4. Why don’t fixed fees affect employee effort?
5. Why might a company offer two different compensation schemes (e.g., commission-only vs. base salary + commission) to its sales staff?

Knowledge Check Answers

1. Marginal cost is the cost of one more unit; average is total cost divided by units.
2. When MR = MC
3. You’re producing too much; reduce output
4. Fixed pay doesn’t change with effort — no additional reward
5. To allow self-selection based on motivation, confidence, and risk preference.

Consultant’s Challenge: Margins that Matter

Scenario:

- You’ve been hired by a meal kit startup struggling to scale profitably. They’re focusing on average cost per box and aren’t sure why they’re still losing money with each new subscriber.

Your Task:

1. What would you say to shift their thinking from average cost to marginal cost and marginal revenue?
2. How could you use marginal analysis to help them make better pricing, production, or effort decisions?

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